FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

# Fluids & Electrolyte Imbalances

Tutorial | Week 2 · Diarrhoea, fluids, HIV & anaemia | Paediatrics | Dr Namugga

Children need fluid prescribed like a drug, by weight (Holliday–Segar 100/50/20 mL/kg/day or 4/2/1 mL/kg/h), as an isotonic fluid with glucose, with boluses only for true shock; sodium must be corrected slowly in both directions, and potassium is given only when the child is passing urine and never as a fast IV push.

**Must know before the ward round**

1. Maintenance (Holliday–Segar): 100 mL/kg/day for the first 10 kg + 50 mL/kg/day for the next 10 kg + 20 mL/kg/day above 20 kg; hourly = 4/2/1 mL/kg/h. UCG 2023 uses 25 mL/kg/day above 20 kg.
2. Use isotonic maintenance fluid (Na 131–154 mmol/L, e.g. 0.9% saline with 5% glucose, or Ringer's lactate with glucose) (NICE NG29); never give plain 5% dextrose as maintenance: it causes hyponatraemia.
3. Sick children (meningitis, pneumonia, bronchiolitis, post-op) retain water (ADH): give 2/3 maintenance (NICE 50–80%) and check sodium daily.
4. Shock (cold hands + CRT > 3 s + weak fast pulse): UCG 20 mL/kg RL/NS as fast as possible; WHO ETAT 2016 10–20 mL/kg over 30–60 min; reassess after each bolus. No rapid bolus if only 1–2 signs (FEAST).
5. SAM with shock: 15 mL/kg over 1 h of a glucose-containing fluid; severe anaemia with shock: blood, not more crystalloid.
6. Deficit (mL) = % dehydration × weight (kg) × 10. Add it to maintenance; most guidelines replace a deficit over 24 h (hypernatraemia over 48 h or more).
7. Neonates (UCG): day 1 60, day 2 90, day 3 120, day 4 onwards 150 mL/kg/day; 10% glucose in the first 2 days.
8. Hyponatraemia with seizures: 2 mL/kg of 2.7–3% saline over 10–15 min (max 100 mL), up to 3 doses (NICE); then raise Na by no more than 8 mmol/L/24 h (NICE max 12).
9. Hypernatraemia: lower Na by ≤ 0.5 mmol/L/h (≤ 10–12 mmol/L in 24 h); fast correction = cerebral oedema and convulsions.
10. Hypokalaemia (K < 3.5): oral KCl 1–2 mmol/kg/dose (max 20 mmol/dose) preferred; IV on the ward 0.2 mmol/kg/h (max 10 mmol/h); max 40 mmol/L in a peripheral line; only if passing urine; check magnesium.
11. Hyperkalaemia with ECG changes: 10% calcium gluconate 0.5 mL/kg IV over 5–10 min (protects the heart), then salbutamol neb 2.5 mg (≤ 25 kg) or 5 mg, insulin 0.1 unit/kg + 10% glucose 5 mL/kg; stop all potassium.
12. Weigh daily, keep a fluid balance chart, and check electrolytes and glucose before IV fluids and at least every 24 h while they run.


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# In a nutshell

Children are **small, wet and fast**: more of their body is water, they turn it over quickly, and they cannot regulate their own intake when they are sick. Errors of a few hundred millilitres, or the wrong type of fluid, cause **hyponatraemic seizures, cerebral oedema, heart failure or death**.

Safe fluid prescribing in children has five habits:

1. **Weigh the child** and calculate by weight.
2. **Decide why** you are giving fluid: resuscitation (bolus), deficit (dehydration), maintenance, or replacement of ongoing losses.
3. **Choose an isotonic fluid** and add **glucose** for maintenance; add **potassium** only once urine is flowing.
4. **Give less** than full maintenance to most sick children (ADH is high).
5. **Monitor**: weight, fluid balance, sodium, potassium and glucose.

> CLINICAL PEARL
> The one thing never to forget: **plain 5% dextrose is not a maintenance fluid.** Once the glucose is used, it is pure water. In a child with meningitis or pneumonia (high ADH), it drops the sodium and causes seizures and brain swelling. Use 0.9% saline (or Ringer's lactate) **with** 5% glucose.

# Definitions & classification

| Term | Definition |
|---|---|
| **Maintenance fluid** | Volume needed to replace normal losses (urine, stool, insensible) in a child who is not eating or drinking |
| **Deficit** | Fluid already lost (dehydration), estimated as % of body weight |
| **Ongoing losses** | Continuing diarrhoea, vomiting, NG aspirates, drains, burns, polyuria; replace volume for volume |
| **Resuscitation (bolus)** | Rapid isotonic fluid to restore circulation in **shock** |
| **Shock (WHO ETAT 2016)** | **All three**: cold extremities, CRT > 3 s, weak and fast pulse |
| **Isotonic fluid** | Sodium 131–154 mmol/L (0.9% saline, Ringer's lactate/Hartmann's, Plasma-Lyte 148) |
| **Hypotonic fluid** | Lower sodium than plasma (0.45% saline, 0.18% saline with 4% glucose, half-strength Darrow's, 5% dextrose) |
| **Hyponatraemia** | Na < 135 mmol/L (moderate 125–129, severe < 125) |
| **Hypernatraemia** | Na > 145 mmol/L (moderate 150–169, severe ≥ 170) |
| **Hypokalaemia** | K < 3.5 mmol/L (mild 3.0–3.4, moderate 2.5–2.9, severe 2.0–2.4, critical < 2.0) |
| **Hyperkalaemia** | K > 5.5 mmol/L (severe ≥ 7, or any level with ECG changes) |

# Epidemiology & Uganda context

- Fluid and electrolyte problems are not a disease but they complicate the **commonest killers** of Ugandan children: **diarrhoea** (dehydration, hypo- and hypernatraemia, hypokalaemia), **severe malaria** (acidosis, AKI,

hyperkalaemia), **meningitis and pneumonia** (SIADH, hyponatraemia), **SAM** (low potassium and magnesium, refeeding), **AGN, HUS and AKI** (overload, hyperkalaemia), **DKA**, **burns**, and **Burkitt lymphoma** (tumour lysis: hyperkalaemia, hyperphosphataemia).

- The **FEAST trial** (2011), run largely in **Uganda (Mulago, Mbale, Soroti)**, Kenya and Tanzania, showed that **fluid boluses increased mortality** in febrile children with impaired perfusion (mostly malaria and sepsis) who were not hypotensive. It changed WHO practice (ETAT 2016).

> **UGANDA CONTEXT**
> On most Ugandan paediatric wards you will find **0.9% saline, Ringer's lactate, 5% and 10% dextrose and 50% dextrose ampoules**, sometimes **half-strength Darrow's in 2.5% dextrose**, and **KCl 15% ampoules** (2 mmol/mL). **Infusion pumps and burettes are scarce**: most children get fluids by gravity through a **20 drops/mL** giving set, which makes overload easy. Electrolytes are available at **Mulago, Mengo, the regional referral hospitals and private labs**, often not at night and rarely at HC IV. **3% saline for IV use** is not routinely stocked outside national/regional referral hospitals. Plan for these limits: use burettes or **micro-drip (60 drops/mL) sets** for small children, write **hourly volumes**, and keep a **fluid chart**.

# Physiology

## Body water

| Age | Total body water (% weight) | Extracellular fluid (% weight) |
|---|---|---|
| Preterm newborn | ~85% | ~50–60% |
| Term newborn | ~75–80% | ~40–45% |
| 1 year | ~60–65% | ~25–30% |
| Adolescent/adult | ~60% | ~20% |

The infant's large ECF is lost first in dehydration, and their turnover is about **3 times** an adult's per kg.

## Where Holliday–Segar comes from

In 1957 Holliday and Segar linked water need to **energy use**: about **100 mL of water per 100 kcal** burned. Energy use per kg falls as children grow, so water per kg falls too: **100 mL/kg/day** for the first 10 kg, **50** for the next 10, **20** for each kg above 20. Daily electrolyte needs are roughly **Na 2–3 mmol/kg/day** and **K 1–2 mmol/kg/day**.

## Why sick children get hyponatraemic

Pain, nausea, fear, surgery, **pneumonia, meningitis, encephalitis, bronchiolitis** and positive-pressure breathing all release **ADH**, independent of plasma tonicity. The kidney retains free water. If the child is also given **hypotonic fluid**, the extra water dilutes the sodium. That is why NICE NG29 (2015, updated 2020) moved to **isotonic maintenance fluids** and why many sick children need **restricted** volumes.

## Sodium and the brain

When plasma sodium falls, water enters brain cells (**cerebral oedema**: headache, vomiting, lethargy, seizures, coma). Within 24–48 h the brain adapts by losing intracellular osmoles. After that adaptation, **rapid correction** of chronic hyponatraemia shrinks the cells and causes **osmotic demyelination**. In **hypernatraemia** the brain makes "idiogenic osmoles"; lowering sodium fast then drives water into brain cells (**cerebral oedema and seizures**). **Both errors come from going too fast.**

### Potassium

About **98% of potassium is inside cells**; serum K is a poor guide to total stores. Insulin, β-agonists and alkalosis drive K **into** cells; acidosis, cell breakdown (haemolysis, tumour lysis, rhabdomyolysis) and insulin deficiency move it **out**. The heart is the target organ: **low K** → flat T waves, U waves, ectopics; **high K** → peaked T waves, wide QRS, sine wave, VF/asystole.

# Clinical acumen

## Focused history

| Ask | Why it matters |
|---|---|
| **Weight** (today and recent, from the child health card) | Every calculation depends on it; weight change = water change |
| **Intake**: breastfeeding, drinking, feeds; what fluids at home | Under-intake, dilute feeds (hyponatraemia) or concentrated formula (hypernatraemia) |
| **Losses**: stools, vomiting (bilious?), urine output, sweating, burns, drains | Size and type of loss decide fluid choice |
| **Urine**: last void, number of wet nappies, colour, polyuria | Oliguria = dehydration or AKI; polyuria with dehydration = DKA, diabetes insipidus, salt-wasting |
| **Neurology**: irritability, lethargy, headache, convulsions | Sodium disorders, hypoglycaemia, meningitis |
| **Weakness, abdominal distension, palpitations** | Hypokalaemia (and hyperkalaemia) |
| **Underlying illness**: malaria, meningitis, pneumonia, kidney disease (puffy face, cola urine), heart disease, SAM, diabetes, cancer on chemotherapy | Predicts the disorder (e.g. AGN → overload, hyperkalaemia) |
| **Medicines**: diuretics, salbutamol, amphotericin, IV fluids already given | Iatrogenic causes; total fluid so far |
| **Nutrition, HIV status, immunisation** | SAM changes all fluid rules; HIV and missed vaccines change the differential |

## Examination

1. **Weight** (same scale), temperature, **HR, RR, BP, $SpO_2$, CRT**.
2. **Hydration**: general condition, eyes, fontanelle, mucosae, tears, skin pinch (doughy in hypernatraemia).
3. **Circulation**: hand temperature, pulse volume, **CRT** (> 3 s abnormal), BP (hypotension is late in children).
4. **Overload**: **puffy eyelids**, pitting oedema, **raised JVP** (older child), **gallop**, basal **crackles**, **tender enlarging liver** (the best sign in infants), rising RR.
5. **Nutrition**: MUAC, oedema of both feet, WLZ.
6. **Neurological**: conscious level (AVPU or Blantyre coma scale), fontanelle, neck stiffness, tone (floppy in hypokalaemia), reflexes.
7. **Abdomen**: distension and absent bowel sounds (ileus, hypokalaemia), masses (Burkitt), bladder.

## Normal vital signs (WHO / ETAT)

| Age | RR (fast if ≥) | HR (awake, normal) | Systolic BP (lower limit approx.) |
|---|---|---|---|
| < 2 months | 60 | 110–160 | 60 |
| 2–12 months | 50 | 100–160 | 70 |
| 1–5 years | 40 | 90–140 | 70 + (2 × age in years) |
| 5–12 years | 30 | 70–120 | 70 + (2 × age), up to 90 |

### Red flags

> RED FLAG
> **Shock**; **convulsions or reduced consciousness** in a child on IV fluids (check glucose and sodium **now**); **Na < 125 or > 160 mmol/L**; **K > 6.5 or < 2.5 mmol/L**, or any K abnormality with **ECG changes**; **no urine for 6–8 h** on fluids; **signs of overload** (gallop, crackles, enlarging liver, puffy eyes); **SAM, severe anaemia, heart disease or AGN** (fluid will kill); and **any child on IV fluids for more than 24 h without an electrolyte check**.

### Thinking like a clinician

- **First ask "is this child in shock?"** Only shock earns a bolus. Fast heart rate from fever, crying or malaria, or cool hands alone, is not shock.
- **If shock, ask "why?"** Hypovolaemia (diarrhoea, bleeding, burns), sepsis, **severe anaemia** (blood, not saline), **SAM** (15 mL/kg slowly), **cardiogenic** (bolus worsens it: big liver, gallop, crackles), DKA (cautious bolus).
- **Calculate, then round down for a sick child.** Meningitis, pneumonia, bronchiolitis, post-op and head injury: **2/3 maintenance**. AGN and AKI: **insensible losses (about 400 mL/m²/day, roughly 15–25 mL/kg/day) plus urine output**.
- **The best monitor is the scale.** A child who gains more than 5% on IV fluids is overloaded.
- **Seizure on IV fluids = glucose, then sodium.** Hypoglycaemia is quicker to fix; hyponatraemia is the commonest hidden cause.
- **Potassium: "no pee, no K"**. Never add KCl until urine is flowing, never inject KCl as a push, and **mix the bag well** after adding it.
- **Common student mistakes**: using plain D5 for maintenance; bolusing a febrile child with malaria who is not shocked; forgetting to subtract oral/NG feeds from the IV volume; writing "1 L 8-hourly" for a 6 kg infant (that is 3 L/day, five times maintenance); correcting chronic hypernatraemia in a few hours.

## Clinical workup

### Step 0: First 5 minutes

1. **ETAT**: airway, breathing ($SpO_2$, give oxygen if < 90%), **circulation** (cold hands, CRT, pulse), coma, convulsions, **severe dehydration**.
2. **Weigh** (or estimate: (age in years + 4) × 2 for 1–10 years, as a last resort).
3. **IV or IO access**; draw **glucose, Hb/HemoCue, electrolytes and creatinine** (if the lab can run them), malaria RDT, group and cross-match if pale.
4. **Glucose < 3 mmol/L → 10% glucose 5 mL/kg IV**.
5. **Shock** → bolus (see Management A), unless SAM (15 mL/kg over 1 h) or severe anaemia (blood) or heart failure (no bolus).
6. **Convulsing** → glucose, benzodiazepine, then think sodium.
7. **Stop** any running hypotonic fluid and any potassium infusion while you assess.

### Step 1: Focused history

- [ ] **Weight and recent weights** → calculations, % dehydration.
- [ ] **Intake and losses** (stool, vomit, urine, burns, drains) → deficit and replacement fluid type.
- [ ] **Fluids already given** (type, volume, time) → avoid double dosing and overload.
- [ ] **Urine output** → AKI and whether potassium is safe.

- [ ] **Underlying diagnosis** (meningitis, pneumonia, AGN, SAM, DKA, malaria, Burkitt, heart disease) → restriction, special protocols.
- [ ] **Neurological symptoms** → urgency of sodium correction.
- [ ] **Medicines** (diuretics, salbutamol, ACE inhibitors, amphotericin, chemotherapy) → K and Na effects.
- [ ] **Feeding** (formula preparation, dilute feeds, breastfeeding) → hypo- or hypernatraemia.
- [ ] **HIV status, nutrition, immunisation** → underlying risk, SAM protocols.

### Step 2: Focused examination

General → vital signs (above) → **hydration** → **circulation** → **signs of overload** → nutrition (MUAC, oedema) → neurology (AVPU/BCS, fontanelle, tone) → abdomen (liver span and edge, bladder, ileus).

| Document these positives | And these negatives |
|---|---|
| Weight and time weighed | "No gallop, no crackles, liver 1 cm" |
| CRT, pulse volume, BP | "No oedema; MUAC 13.5 cm" |
| Hydration signs | "Passed urine at 14:00" |
| Conscious level (BCS/AVPU) | "Fontanelle not bulging; no neck stiffness" |

### Step 3: Bedside tests

| Test | How | What it tells you |
|---|---|---|
| **Weight** | Same scale daily (or 12-hourly) | Fluid gain/loss: 1 kg ≈ 1 L |
| **RBS** | Glucometer | Hypo- or hyperglycaemia (DKA); glucose in fluids needed? |
| **Urine output** | Weigh nappies (1 g = 1 mL) or bag | Normal ≥ **1 mL/kg/h** (≥ 2 mL/kg/h in infants); < 0.5 mL/kg/h = oliguria |
| **Urine dipstick and SG** | Dipstick | SG > 1.020 concentrated (dehydration, SIADH); glucose and ketones (DKA); blood and protein (AGN) |
| **HemoCue Hb** | Finger prick | Severe anaemia changes the shock plan |
| **ECG / cardiac monitor** | 3-lead monitor or 12-lead | Peaked T waves, wide QRS (high K); flat T, U waves (low K) |
| **Point-of-care electrolytes / blood gas** | i-STAT or ward gas (if available) | Na, K, glucose, lactate, pH, bicarbonate in minutes |
| **Fluid balance chart** | Hourly input and output | Running total guides the next prescription |

### Step 4: Laboratory & imaging

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **Serum Na, K, Cl, bicarbonate** | Before IV fluids (except short elective cases), then at least **every 24 h**; **4–6 h** after starting if very sick or abnormal | Na < 135 or > 145; K < 3.5 or > 5.5; low bicarbonate | Choice of fluid, rate of correction, potassium dose. **RRH/NRH, Mengo, private labs**; rarely at HC IV |
| **Urea and creatinine** | Dehydration, oliguria, AGN, malaria with dark urine, HUS | Raised (pre-renal or intrinsic AKI) | Fluid restriction, dialysis referral (peritoneal dialysis at Mulago) |

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **Serum and urine osmolality, urine sodium** | Unexplained hyponatraemia | Urine Na > 20–30 mmol/L with concentrated urine = SIADH or renal salt loss; < 20 = extrarenal loss | Restrict (SIADH) vs replace salt (salt-wasting) |
| **Blood gas / lactate** | Shock, DKA, severe malaria, sepsis | pH < 7.3, bicarbonate < 15, lactate > 4 | Severity; response to treatment. NRH/ICU |
| **Magnesium, calcium, phosphate** | Refractory hypokalaemia, SAM refeeding, tumour lysis, tetany | Low Mg; low Ca; low $PO_4$ (refeeding) or high $PO_4$ (tumour lysis) | Replace Mg before K corrects; slow feeds |
| **Glucose (lab)** | DKA, hypoglycaemia | > 11 or < 3 mmol/L | Insulin protocol or glucose |
| **FBC** | Pallor, sepsis | Severe anaemia; haemoconcentration | Transfuse; sepsis care |
| **Chest X-ray** | Suspected overload or heart failure | Cardiomegaly, pulmonary oedema | Stop fluids, furosemide |

## Step 5: Putting it together

### Model one-line summary

> "A 3-year-old boy, 13 kg, day 2 of treatment for bacterial meningitis, on 'D5 1 L 8-hourly' since admission, who had a generalised seizure this morning; Na 121 mmol/L, glucose 6.4 mmol/L: acute symptomatic hyponatraemia from hypotonic fluid plus SIADH, with fluid overload (3 L/day against a maintenance of 1150 mL)."

### Problem list

1. Acute symptomatic hyponatraemia (Na 121) with seizure.
2. Iatrogenic fluid overload with a hypotonic fluid.
3. Bacterial meningitis (SIADH risk) on ceftriaxone.

**Working diagnosis and plan**: stop D5; **2 mL/kg 3% saline (26 mL) over 10–15 min**, repeat once if still seizing; then **isotonic fluid with glucose at 2/3 maintenance**; Na **2–4 hourly**; aim for a rise of **no more than 8 mmol/L in 24 h** after symptoms stop.

EXAM FAVOURITE

**Worked example 1: Holliday–Segar.**

- **7 kg** infant: 7 × 100 = **700 mL/day** = 29 mL/h (or 7 × 4 = 28 mL/h).
- **15 kg** child: (10 × 100) + (5 × 50) = 1000 + 250 = **1250 mL/day** = 52 mL/h (or 40 + 10 = 50 mL/h).
- **32 kg** child: 1000 + 500 + (12 × 20) = **1740 mL/day** = 72 mL/h (or 40 + 20 + 12 = 72 mL/h). **UCG 2023 variant** (25 mL/kg above 20 kg): 32 kg = 1000 + 500 + 300 = 1800 mL/day. **2/3 maintenance** for the 15 kg child with meningitis = 1250 × 2/3 ≈ **830 mL/day ≈ 35 mL/h**.

EXAM FAVOURITE

**Worked example 2: deficit plus maintenance, and drops per minute.** A 12 kg child is 7% dehydrated, cannot drink and needs IV fluids (not shocked).

- Deficit = 7 × 12 × 10 = **840 mL**; maintenance = 1000 + 100 = **1100 mL**; total **1940 mL over 24 h ≈ 81 mL/h**.
- With a **20 drops/mL** giving set: drops/min = mL/h × 20 ÷ 60 = 81 ÷ 3 ≈ **27 drops/min**.
- With a **60 drops/mL** micro-drip/burette: drops/min = mL/h = **81 drops/min**.
- Add **KCl 20 mmol/L** once he passes urine (15% KCl = 2 mmol/mL → **10 mL into 1 L**, or 5 mL into a 500 mL bag; mix well).
- Weigh again in 12 h; recheck Na and K at 24 h. (For diarrhoeal dehydration use the WHO plans instead; see the Diarrhoea topic.)

# Differential diagnosis

Causes of each disorder in Ugandan children:

| Disorder | Common causes | Distinguishing features | Key investigation |
|---|---|---|---|
| **Hypovolaemia** | Diarrhoea and vomiting, cholera, burns, haemorrhage, DKA, sepsis (capillary leak) | Dehydration signs, low urine output, weight loss | Weight, urea, Na |
| **Fluid overload** | Excess IV fluid, **AGN**, AKI, heart failure (rheumatic, congenital), severe anaemia, SAM refeeding | Puffy eyes, oedema, big liver, gallop, crackles, weight gain | CXR, creatinine, urine |
| **Hyponatraemia, hypovolaemic** | Diarrhoea replaced with water or dilute fluids, salt-wasting CAH, diuretics, cerebral salt wasting | Dehydrated; high K suggests **CAH or adrenal crisis** | Urine Na, K, glucose, 17-OHP (CAH) |
| **Hyponatraemia, euvolaemic** | **SIADH** (meningitis, pneumonia, TB meningitis, encephalitis, post-op), **hypotonic IV fluids**, water intoxication (diluted formula) | Normal hydration, concentrated urine, urine Na high | Urine and serum osmolality, urine Na |
| **Hyponatraemia, hypervolaemic** | **Nephrotic syndrome**, heart failure, cirrhosis, AKI | Oedema | Urine protein, albumin, echo |
| **Hypernatraemia** | Diarrhoea with high fever, concentrated formula, poor breastfeeding in a newborn, salt poisoning, diabetes insipidus (after meningitis or head injury), excess sodium bicarbonate | Very irritable, doughy skin, thirst; polyuria in DI | Urine osmolality, Na |
| **Hypokalaemia** | **Diarrhoea**, vomiting (pyloric stenosis: low Cl, alkalosis), **SAM**, refeeding, **DKA treatment**, salbutamol, diuretics, amphotericin, hypomagnesaemia | Floppy, ileus, ECG changes | K, Mg, bicarbonate |
| **Hyperkalaemia** | **AKI** (AGN, HUS, malaria, dehydration), **haemolysed sample** (heel prick), **tumour lysis (Burkitt)**, haemolysis, **CAH**, rhabdomyolysis, massive transfusion, K in fluids, ACE inhibitors | Oliguria, ECG changes | Repeat free-flowing sample, creatinine, ECG |

# Investigations

| Test | What you are looking for | Why |
|---|---|---|
| Electrolytes, urea, creatinine | Na, K, AKI | Choose fluid and rate; detect harm |
| Glucose | Hypo/hyperglycaemia | Maintenance glucose; DKA |
| Urine output and SG | Oliguria, concentration | Perfusion, SIADH, AKI |
| Blood gas | Acidosis, lactate | Severity of shock and DKA |
| Mg, Ca, $PO_4$ | Associated deficits | Refractory hypokalaemia, refeeding |
| ECG | K-related changes | Urgency of K treatment |
| Daily weight | Gain or loss | Overload or under-replacement |

# Management

### A. Resuscitation: is the child in shock?

**Shock = cold hands + CRT > 3 s + weak, fast pulse** (all three; WHO ETAT 2016).

| Situation | Fluid | Volume and rate | Then |
|---|---|---|---|
| **Shock, not malnourished, not anaemic** | **Ringer's lactate or 0.9% saline** (never glucose-only) | **UCG 2023: 20 mL/kg as rapidly as possible**; WHO ETAT 2016: **10–20 mL/kg over 30–60 min**; NICE NG29: 10 mL/kg over < 10 min | Reassess after each bolus; repeat 10–20 mL/kg if still shocked; involve a senior after 20 mL/kg; consider inotropes (adrenaline/dopamine) and septic shock after 40–60 mL/kg |
| **Impaired circulation (1–2 signs only)**, febrile illness, severe malaria, pneumonia, meningitis | **No bolus** (FEAST) | **Maintenance** fluids | Reassess within 1 hour |
| **Severe dehydration from diarrhoea** | Ringer's lactate | **WHO Plan C**: 30 mL/kg then 70 mL/kg (see Diarrhoea topic) | Reassess, ORS |
| **SAM with shock** | RL with 5% glucose, half-strength Darrow's with 5% glucose, or 0.45% saline with 5% glucose | **15 mL/kg over 1 hour**, repeat once if improving | Oral/NG ReSoMal; if no improvement, **blood 10 mL/kg over 3 h** |
| **Shock with severe anaemia** (Hb < 5 g/dL) | **Whole blood 20 mL/kg** (or packed cells 10 mL/kg) | As early as possible | Other IV fluids only to maintain hydration |
| **Haemorrhage** | Blood (UCG: 20 mL/kg) after initial crystalloid | | Stop the bleeding |
| **DKA with shock** | 0.9% saline | ISPAD 2022: **10–20 mL/kg over 20–30 min** (repeat if needed) | Deficit over 24–48 h, insulin after the first hour |
| **Cardiogenic shock / heart failure** | **No bolus**; small cautious 5–10 mL/kg only on senior advice | | Oxygen, furosemide, inotropes |

RED FLAG

**Stop the bolus** if you hear **crackles**, a **gallop**, the **liver edge moves down**, the eyelids swell, or the child becomes more breathless or less conscious. **Never** give glucose-only fluids as a bolus (hyperglycaemia and hyponatraemia).

## B. Maintenance fluids

**Volume (Holliday–Segar, NICE NG29, WHO Pocket Book):**

| Weight | Per day | Per hour ("4-2-1") |
|---|---|---|
| First 10 kg | **100 mL/kg** | **4 mL/kg/h** |
| Next 10 kg (11–20 kg) | **+ 50 mL/kg** | **+ 2 mL/kg/h** |
| Each kg above 20 kg | **+ 20 mL/kg** (UCG: 25 mL/kg) | **+ 1 mL/kg/h** |
| Cap | About **2–2.5 L/day** (adult maintenance) | |

**Adjustments:**

- **Fever**: + about **10% per °C** above 37 °C (UCG).
- **Restrict to 2/3 (NICE: 50–80%)** in conditions with high ADH: **meningitis**, **pneumonia**, **bronchiolitis**, **head injury**, **post-operative**, encephalitis, ventilated children.
- **AGN / AKI / oliguria**: insensible losses (**about 400 mL/m²/day**, roughly **15–25 mL/kg/day** in small children) **plus the previous day's urine output**.
- **Heart failure**: 50–70% of maintenance with furosemide.
- **Subtract** oral, NG and drug volumes from the IV total.
- Change to the **oral route** as soon as the child can drink or take NG feeds.

**Neonates (UCG 2023):**

| Day of life | Total fluid (oral and/or IV) |
|---|---|
| Day 1 | **60 mL/kg/day** of 10% glucose |
| Day 2 | **90 mL/kg/day** of 10% glucose |
| Day 3 | **120 mL/kg/day** (half-normal saline with 5% glucose, or 10% glucose with added sodium per local protocol) |
| Day 4 onwards | **150 mL/kg/day**; up to **180 mL/kg/day** once oral feeding is established |

*Do not exceed 100 mL/kg/day as IV fluid alone unless the baby is dehydrated or under phototherapy or a radiant heater (UCG). Breastfeed or give expressed breast milk by NG tube whenever possible.*

**Type of maintenance fluid:**

| Child | Fluid |
|---|---|
| **1 month and older** (NICE NG29, RCH) | **Isotonic with glucose**: **0.9% saline with 5% glucose** (± KCl 20 mmol/L) or **Ringer's lactate with 5% glucose**; Plasma-Lyte 148 with 5% glucose where available |
| **Term neonate ≥ 8 days** | Isotonic crystalloid with **5–10% glucose** |
| **Neonate first days** | **10% glucose** (sodium added from day 2–3 when urine output is established) |
| **SAM** | **Avoid IV maintenance**; feed F-75 orally/NG |
| **Older UCG practice** | One-third 0.9% saline and two-thirds 5% dextrose (adult chapter) or half-strength Darrow's in 2.5% glucose: **hypotonic**; check sodium daily if used |

UGANDA CONTEXT

**Making "0.9% saline with 5% glucose" on a Ugandan ward**: remove **50 mL** from a **500 mL bag of 0.9% saline** and add **50 mL of 50% dextrose** → about **5% glucose in 0.9% saline**. **10% glucose** from stock: **1 part 50% + 4 parts water**, or **1 part 50% + 9 parts 5% dextrose** (UCG). Label the bag clearly with what you added, the time and your name.

## C. Deficit and ongoing losses

1. **Deficit (mL) = % dehydration × weight (kg) × 10.** (5% of 10 kg = 500 mL.)
2. Replace the deficit **over 24 h** in most children (RCH: no more than about **5% deficit replaced IV in 24 h**, then reassess); **over 48–72 h** in hypernatraemia; diarrhoeal dehydration uses the **WHO plans** instead.
3. **Ongoing losses**: measure and replace **volume for volume** every 4–6 h: diarrhoea with **ORS** or Ringer's lactate; **NG/vomit losses with 0.9% saline + 10–20 mmol/L KCl**; burns per a burns formula (e.g. Parkland 3–4 mL/kg/% burn in 24 h plus maintenance).

## D. Composition of common fluids (mmol/L)

| Fluid | Na | K | Cl | Other | Glucose | Use |
|---|---|---|---|---|---|---|
| **0.9% saline** ("normal saline") | 154 | 0 | 154 | | 0 | Bolus, deficit, maintenance base |
| **Ringer's lactate / Hartmann's** | 130–131 | 4–5 | 109–111 | Lactate 28–29, Ca 2 | 0 | **First choice for resuscitation and Plan C** |
| **Plasma-Lyte 148** | 140 | 5 | 98 | Mg 1.5, acetate, gluconate | 0 | Bolus/maintenance (private/NRH) |
| **5% dextrose** | 0 | 0 | 0 | | 50 g/L | **Not for maintenance alone**; vehicle for drugs |
| **10% dextrose** | 0 | 0 | 0 | | 100 g/L | Hypoglycaemia (5 mL/kg), neonates |
| **0.45% saline + 5% dextrose** | 77 | 0 | 77 | | 50 g/L | Hypotonic; slow phase of hypernatraemia; SAM shock |
| **Half-strength Darrow's + 2.5% dextrose** | 61 | 17 | 52 | Lactate 27 | 25 g/L | Old UCG option; hypotonic |
| **Low-osmolarity ORS** | 75 | 20 | 65 | Citrate 10 | 75 mmol/L | Oral rehydration |
| **ReSoMal** | 45 | 40 | | Mg, Zn, Cu | | SAM rehydration |
| **15% KCl ampoule** | | **2 mmol/mL** | | | | **Additive only; never undiluted** |

## E. Hyponatraemia (Na < 135 mmol/L)

**Step 1: Is the child symptomatic?** (Headache, vomiting, lethargy, confusion, **seizures**, reduced consciousness.)

**Acute symptomatic hyponatraemia (emergency):**

1. ABC; oxygen; treat seizure (benzodiazepine) and check **glucose**.
2. **Hypertonic saline: 2 mL/kg of 2.7% (or 3%) NaCl IV over 10–15 min (max 100 mL)** (NICE NG29). Repeat once if symptoms persist; check Na and consider a **third dose** if still symptomatic. Each 1 mL/kg of 3%

raises Na by about **1 mmol/L**; a rise of **4–6 mmol/L** usually stops seizures.

3. **Stop hypotonic fluids**; continue **isotonic** fluid at a **restricted** rate.
4. Once symptoms stop: total rise **≤ 8 mmol/L in 24 h** (Medscape, RCH 6–8; **NICE: never more than 12 mmol/L in 24 h**). Check Na every **2–4 hours**.

**Asymptomatic or mild:**

| Cause (volume state) | Treatment |
|---|---|
| **Hypovolaemic** (diarrhoea, salt loss) | Rehydrate with **isotonic fluid** (ORS, RL or 0.9% saline); Na rises as volume is restored. Watch for an overshoot once ADH switches off (brisk urine) |
| **Euvolaemic / SIADH** (meningitis, pneumonia, post-op, hypotonic fluids) | **Restrict** to 50–2/3 maintenance; **isotonic** fluid only; treat the cause. **Do not manage acute hyponatraemic encephalopathy by fluid restriction alone** (NICE) |
| **Hypervolaemic** (nephrotic, heart failure, AKI) | Restrict fluid and salt; diuretics (furosemide 1 mg/kg) for overload; treat the cause |
| **Salt-wasting CAH / adrenal crisis** (hyponatraemia + **hyperkalaemia** + hypoglycaemia in an infant; ambiguous genitalia in girls) | 0.9% saline 20 mL/kg if shocked, **hydrocortisone**, glucose; specialist referral |

EXAM FAVOURITE

**Worked example 3: hyponatraemia.** A 10 kg child with pneumonia, on "D5 500 mL 6-hourly" (2 L/day; maintenance 1 L), seizes. Na **118 mmol/L**, glucose 7.

- Give **3% saline 20 mL (2 mL/kg) over 10–15 min**; seizure stops; repeat Na 30 min later = **122**.
- Change to **0.9% saline with 5% glucose at 2/3 maintenance = 660 mL/day ≈ 28 mL/h**.
- Target Na at 24 h: no higher than **118 + 8 = 126 mmol/L**. If it rises faster (brisk urine as ADH falls), slow the saline and give some 5% glucose on senior advice.

## F. Hypernatraemia (Na > 145 mmol/L)

1. **Treat shock first** with **0.9% saline or RL boluses** (isotonic fluid is relatively hypotonic to this child's plasma).
2. Then correct **slowly**: sodium should fall **no faster than 0.5 mmol/L per hour** (about **10–12 mmol/L in 24 h**; NICE: never more than 12 mmol/L in 24 h). Replace the deficit over **48 hours or more** (72 h if Na > 160–170).
3. **Fluid**: **ORS by mouth or NG** where possible (NICE: ORS is also suitable in hypernatraemia). IV: start with **0.9% saline (with glucose)**; if sodium does not fall, move to **0.45% saline with 5% glucose** (NICE algorithm).
4. **Check Na every 2–4 hours** at first (RCH: 1–2 h after starting, then 4–6-hourly). **If Na falls > 0.5 mmol/L/h, slow or stop the fluid.**
5. Add **KCl 20–40 mmol/L** once the child passes urine (helps water enter cells; Medscape).
6. **Seizures during correction** = cerebral oedema: stop hypotonic fluid, give **3% saline 2–3 mL/kg** (or mannitol) and call ICU.
7. **Fix the cause**: correct formula preparation (1 level scoop per 30 mL water), support breastfeeding in a newborn (weigh, observe a feed), treat fever, test urine for **diabetes insipidus** (dilute urine despite hypernatraemia).

EXAM FAVOURITE

**Worked example 4: free-water deficit (rough guide).** Free water deficit (L) = 0.6 × weight × (current Na ÷ 145 − 1). A 6 kg infant with Na 166: 0.6 × 6 × (166/145 − 1) = 3.6 × 0.145 ≈ **0.52 L**. Spread over **48 h** with maintenance (600 mL/day × 2 = 1.2 L) → about **1.7 L in 48 h ≈ 36 mL/h**, then **measure Na** and adjust. Formulas only start you off; **serial sodium values drive the rate**.

## G. Hypokalaemia (K < 3.5 mmol/L)

1. **Confirm** (free-flowing venous sample), check **Mg**, bicarbonate, glucose; **ECG** if K < 3.0 or symptoms.
2. **Ensure the child is passing urine.**
3. **Oral route preferred** (RCH): **KCl 1–2 mmol/kg/dose (max 20 mmol/dose)**; repeat after checking K, to a **maximum 5 mmol/kg/day (max 50 mmol/day)**. Maintenance if needed **2–5 mmol/kg/day** in divided doses. Slow-K tablet = 8 mmol; give with food.
4. **IV** if unable to take oral, K < 2.5 or ECG changes:
   - **Add KCl to maintenance fluid: 20 mmol/L** (10 mL of 15% KCl per 1 L), up to **40 mmol/L** in a peripheral line.
   - **Acute replacement on a general ward: 0.2 mmol/kg/h for 3 hours (max 10 mmol/h)**, ECG monitoring if K < 3; repeat K 1 hour after.
   - **Higher rates (0.4 mmol/kg/h, max 20 mmol/h)** only in **ICU** via a central line with continuous monitoring.
   - **Never** give KCl as an IV push or IM. **Invert the bag several times** after adding KCl.
5. **Correct magnesium** if low (hypokalaemia is refractory until Mg is replaced).
6. **SAM**: extra potassium **3–4 mmol/kg/day** and magnesium **0.4–0.6 mmol/kg/day**, given in feeds or ReSoMal (the combined electrolyte/mineral mix), not IV.

## H. Hyperkalaemia (K > 5.5 mmol/L)

1. **Is it real?** Repeat a **free-flowing venous sample** (heel pricks and squeezed fingers haemolyse); but **do not delay treatment** if ECG changes or K ≥ 7.
2. **Stop all potassium** (IV additives, oral supplements, RL is low in K but stop it in severe AKI), stop ACE inhibitors and K-sparing diuretics. **Cardiac monitor and ECG.**
3. **Treat in order:**

| Step | Drug | Dose | Route / rate | Onset / duration | Notes |
|---|---|---|---|---|---|
| **1. Protect the heart** (ECG changes or K ≥ 7) | **Calcium gluconate 10%** | **0.5 mL/kg (max 20 mL)**; RCH 0.68 mL/kg (0.15 mmol/kg), max 30 mL | IV/IO over **2–5 min** if unstable, 15–20 min if stable | < 3 min; lasts ~30 min | Does **not** lower K; repeat if ECG still abnormal; not in the same line as bicarbonate; avoid if on digoxin |
| **2. Shift K into cells** | **Salbutamol** | **2.5 mg (≤ 25 kg); 5 mg (> 25 kg)** | Nebulised, repeat 1–2-hourly | 30 min; 2–3 h | Quick, available everywhere in Uganda |
| | **Insulin (soluble/Actrapid) + glucose** | **Insulin 0.1 unit/kg IV (max 10 units) + 10% glucose 5 mL/kg IV** | Bolus, then infusion if needed | 15 min; 2–3 h | **Check glucose every 30–60 min** (hypoglycaemia risk) |
| | **Sodium bicarbonate 8.4%** | **1 mmol/kg (1 mL/kg)** | Slow IV over 30 min (dilute 1:1 in neonates and infants) | 30–60 min | **Only if metabolic acidosis**; not with calcium |

| Step | Drug | Dose | Route / rate | Onset / duration | Notes |
|---|---|---|---|---|---|
| **3. Remove K from the body** | **Calcium/sodium polystyrene sulfonate (Resonium)** | **0.5–1 g/kg (max 30 g)** | Oral or rectal, 6-hourly | Hours | Mild effect; constipation |
| | **Furosemide** | **1 mg/kg** (max 40 mg) | IV | | Only if kidneys produce urine |
| | **Dialysis** | | Peritoneal or haemodialysis | | Refractory hyperkalaemia, AKI: **Mulago (PD), some RRHs** |

4. **Treat the cause**: AKI (AGN, HUS, malaria), tumour lysis (hydration, allopurinol, avoid K), adrenal crisis (hydrocortisone), haemolysis.

DRUG DOSING

**Potassium safety rules**: **15% KCl = 2 mmol/mL**; **10 mL into 1 L = 20 mmol/L**; **maximum 40 mmol/L peripherally**; **maximum 0.2 mmol/kg/h on a general ward (max 10 mmol/h)**; **never undiluted, never IV push, never IM**; only once urine output is established. Store KCl ampoules separately from saline and water ampoules: look-alike ampoules have killed children.

### I. Monitoring every child on IV fluids

1. **Prescribe in mL/h** with the fluid type, additives and duration. Review the prescription **at least every 12–24 h**.
2. **Fluid balance chart**: all input (IV, oral, NG, drugs) and output (urine, stool, vomit, drains) every hour; running totals.
3. **Weigh daily** (12-hourly in infants, SAM, AGN).
4. **Electrolytes and glucose** before starting (except most elective surgery) and **at least every 24 h**; every **4–6 h** if sick, abnormal or correcting sodium.
5. **Clinical review** at least every 4 h: hydration, overload signs, conscious level, urine output.
6. **Switch to oral** as soon as possible, and take the line out.

### J. Setting-specific plan

| Setting | What you can realistically do |
|---|---|
| **HC III** | Weigh; ETAT; IV/IO access; RL/NS boluses; WHO Plans; 10% glucose; NG ORS; refer for electrolyte disorders |
| **HC IV** | Inpatient IV maintenance with glucose; KCl additive; HemoCue; transfusion where a blood fridge exists; **no electrolyte lab in most**: treat empirically by protocol and refer if seizures or AKI |
| **RRH / Mengo / Mulago** | Electrolytes, creatinine, ECG, insulin and calcium for hyperkalaemia, 3% saline (NRH/RRH), peritoneal dialysis (Mulago), HDU/ICU |

## Complications

| Error | Consequence |
|---|---|
| **Too much fluid** | Pulmonary oedema, heart failure, raised ICP, death (FEAST), worse outcomes in SAM, AGN, severe anaemia |
| **Hypotonic fluid** | **Hyponatraemic encephalopathy**: seizures, brainstem herniation, death |

| Error | Consequence |
|---|---|
| **Correcting hyponatraemia too fast** | **Osmotic demyelination** (weeks later: quadriparesis, mutism) |
| **Correcting hypernatraemia too fast** | **Cerebral oedema**, seizures, death |
| **Too little fluid** | Persistent shock, AKI, cerebral venous thrombosis (hypernatraemia), death |
| **Potassium errors** | Cardiac arrest (KCl push or overdose), arrhythmias, ileus, paralysis |
| **Glucose errors** | Hypoglycaemia (no glucose in maintenance; insulin), hyperglycaemia (glucose boluses) |
| **IV line** | Thrombophlebitis, extravasation injury (KCl, calcium, dextrose 10%+), line sepsis |

# Ward-round & exam pearls

**Model case presentation:**

> "Sarah is a **3-year-old girl, 13 kg** (WAZ −0.8, WHZ −0.5, MUAC 15 cm, no oedema), **fully immunised** on the UNEPI schedule, **HIV-negative**, eating family foods, admitted 2 days ago with **bacterial meningitis** on ceftriaxone. Since admission she has been on **5% dextrose, 500 mL 6-hourly (2 L/day)**. This morning she had a **generalised tonic-clonic seizure** for 4 minutes, stopped with rectal diazepam. Now **BCS 4/5**, HR 128, RR 32, BP 98/60, CRT 2 s, **puffy eyelids**, liver 2 cm, no crackles. RBS 6.4. **Na 121, K 3.4**, creatinine normal. **Summary**: acute symptomatic hyponatraemia from a **hypotonic fluid at nearly twice maintenance** on top of **SIADH from meningitis**. Plan: **3% saline 26 mL over 15 min**, stop D5, **0.9% saline with 5% glucose + KCl 20 mmol/L at 2/3 maintenance (35 mL/h)**, Na 4-hourly, aiming for **≤ 8 mmol/L rise in 24 h**, daily weight, fluid chart, and an incident report."

**Likely viva questions (with answers)**

1. *Calculate maintenance for a 24 kg child.* 1000 + 500 + 80 = 1580 mL/day (≈ 66 mL/h); UCG variant 1600 mL/day.
2. *Why isotonic maintenance fluid?* Sick children have high ADH; hypotonic fluids cause hyponatraemia and encephalopathy (NICE NG29).
3. *What did FEAST show?* Boluses (20–40 mL/kg) of saline or albumin increased 48-h mortality in African children with severe febrile illness and impaired perfusion; WHO now restricts boluses to children with all three signs of shock.
4. *Name three causes of hyperkalaemia in Ugandan children.* AKI (AGN, HUS, malaria), tumour lysis in Burkitt lymphoma, haemolysed samples; also CAH, transfusion.
5. *First drug for hyperkalaemia with peaked T waves?* IV calcium gluconate 10% 0.5 mL/kg over 2–5 min; it stabilises the myocardium but does not lower K.
6. *How fast may you correct hypernatraemia?* No more than 0.5 mmol/L/h, 10–12 mmol/L per 24 h, over 48 h or more.
7. *What is the maximum potassium concentration in a peripheral IV?* 40 mmol/L; maximum rate 0.2 mmol/kg/h on a ward.

**OSCE tips**: for "write an IV fluid prescription", state the **weight**, show the **calculation**, name the **fluid with glucose**, add **KCl only when passing urine**, give **mL/h and drops/min** for the giving set, and write the **monitoring** (weight, chart, Na/K/glucose at 24 h). For "interpret these electrolytes", **comment on the volume state first**, then sodium, potassium, bicarbonate, urea/creatinine, glucose, and link each to the clinical story.

# Mnemonics & memory aids

- **"4-2-1"** for hourly maintenance; **"100-50-20"** per day.
- **"Deficit = % × kg × 10"**.
- **"No pee, no K"**; **"K max 40 (mmol/L) and 0.2 (mmol/kg/h)"**.
- **"Sodium: slow in, slow out"** (≤ 8–12 up, ≤ 10–12 down per 24 h; 0.5 mmol/L/h).
- **Hyperkalaemia "C BIG K Drop"**: **C**alcium, **B**icarbonate (if acidotic), **I**nsulin + **G**lucose, **K**ayexalate/Resonium, **D**iuretic/**D**ialysis (and salbutamol nebs).
- **"Shock gets a bolus; sick gets 2/3; SAM gets 15; anaemia gets blood."**
- **Drops/min with a 20-drop set = mL/h ÷ 3**; with a 60-drop micro-drip = **mL/h**.

# High-yield summary

### Numbers & doses to know

| Item | Value |
|---|---|
| Maintenance (Holliday–Segar) | **100 / 50 / 20 mL/kg/day** (UCG: 25 above 20 kg); **4 / 2 / 1 mL/kg/h** |
| Fever | + **10% per °C** |
| Restricted maintenance (high ADH) | **2/3** (NICE **50–80%**) |
| Neonatal fluids (UCG) | Day 1 **60**, day 2 **90**, day 3 **120**, day 4+ **150 mL/kg/day** (oral up to 180) |
| Shock bolus | UCG **20 mL/kg RL/NS rapidly**; WHO ETAT **10–20 mL/kg over 30–60 min**; NICE **10 mL/kg < 10 min** |
| SAM shock | **15 mL/kg over 1 h**, glucose-containing fluid |
| Shock + severe anaemia | **Blood** (whole blood 20 mL/kg or packed cells 10 mL/kg) |
| Deficit | **% × kg × 10 mL** |
| Normal urine output | ≥ **1 mL/kg/h** (infants ≥ 2) |
| Hypoglycaemia | **10% glucose 5 mL/kg** IV |
| Symptomatic hyponatraemia | **2 mL/kg of 2.7–3% saline over 10–15 min** (max 100 mL), up to 3 doses |
| Na correction limits | Up: **≤ 8 mmol/L/24 h** (NICE max 12). Down: **≤ 0.5 mmol/L/h** (≤ 10–12/24 h) |
| Oral KCl | **1–2 mmol/kg/dose** (max 20 mmol/dose; max 5 mmol/kg/day) |
| IV KCl | **20 mmol/L** in maintenance (max **40 mmol/L** peripheral); acute **0.2 mmol/kg/h** (max 10 mmol/h) |
| 15% KCl | **2 mmol/mL** |
| Calcium gluconate 10% | **0.5 mL/kg** (max 20 mL; RCH max 30 mL) over 2–5 min |
| Salbutamol (hyperkalaemia) | **2.5 mg** (≤ 25 kg) / **5 mg** neb |
| Insulin + glucose | **0.1 unit/kg** + **10% glucose 5 mL/kg** |
| Sodium bicarbonate 8.4% | **1 mmol/kg (1 mL/kg)** over 30 min, only if acidotic |
| Resonium | **0.5–1 g/kg** 6-hourly (max 30 g) |
| Furosemide | **1 mg/kg** IV |

## Classic exam traps

- **"5% dextrose for maintenance."** Wrong: hypotonic → hyponatraemia; use 0.9% saline (or RL) **with** 5% glucose.
- **"Febrile child with malaria and cold hands → 20 mL/kg bolus."** Wrong unless all 3 signs of shock (FEAST).
- **"Bolus for a child with severe anaemia in shock."** Give **blood**.
- **"Correct Na 116 to 135 today."** Wrong: ≤ 8 mmol/L in 24 h after symptoms settle.
- **"Calcium gluconate lowers potassium."** Wrong: it protects the heart; insulin/glucose and salbutamol lower K.
- **"Add KCl to the first bolus."** Wrong: never in a bolus; only in maintenance once urine flows.
- **"K 6.8 from a heel prick in a well baby → insulin now."** Repeat a free-flowing sample first (if ECG normal).
- **"Fluid-restrict a child with hyponatraemic seizures."** Wrong: give hypertonic saline first.

## Questions seniors ask

**Write maintenance for this 18 kg child with pneumonia.** Holliday–Segar 1000 + 400 = 1400 mL/day; pneumonia (SIADH risk) → 2/3 ≈ 930 mL/day ≈ 39 mL/h of 0.9% saline with 5% glucose, KCl 20 mmol/L once passing urine; weigh daily, Na/K/glucose at 24 h; switch to oral feeds as soon as possible.

**Why not bolus every child with a fast pulse and cool hands?** FEAST (East Africa, 2011) showed increased mortality; only children with all three signs of shock get boluses, and SAM, severe anaemia and heart failure have special rules.

**This child on IV fluids is drowsy. What do you check first?** Glucose (bedside), then sodium; look at the fluid chart (type and volume); look for meningitis or raised ICP.

**How do you know the child is overloaded?** Weight gain, puffy eyelids, oedema, enlarging tender liver, gallop, crackles, rising RR and falling $SpO_2$, raised JVP in older children.

**The K is 7.2 with peaked T waves. What do you do in the next 10 minutes?** Stop all K, monitor, calcium gluconate 10% 0.5 mL/kg IV over 2–5 min, salbutamol neb, insulin 0.1 unit/kg with 10% glucose 5 mL/kg, bicarbonate if acidotic, repeat K and ECG, then Resonium, furosemide if passing urine, and call for dialysis if refractory.

**How would you make 10% glucose if only 50% and 5% are available?** One part 50% to 9 parts 5% (e.g. 10 mL of 50% + 90 mL of 5%), or one part 50% to 4 parts sterile water.

SELF-TEST

# Flashcards

28 cards. Cover the right-hand column and test yourself.

| # | Question | Answer |
|---|---|---|
| 1 | Holliday–Segar maintenance per day | **100 mL/kg** for the first 10 kg + **50 mL/kg** for the next 10 kg + **20 mL/kg** for each kg above 20 kg (UCG 2023 uses 25 mL/kg above 20 kg). |
| 2 | Holliday–Segar per hour ('4-2-1') | **4 mL/kg/h** for the first 10 kg, **2 mL/kg/h** for the next 10 kg, **1 mL/kg/h** above 20 kg. |
| 3 | Maintenance for a 15 kg child | 1000 + 250 = **1250 mL/day ≈ 52 mL/h**. |
| 4 | Which fluid for IV maintenance in children over 1 month (NICE NG29)? | **Isotonic** crystalloid (Na 131–154 mmol/L) **with glucose**, e.g. **0.9% saline + 5% glucose** (± KCl 20 mmol/L). Not plain 5% dextrose. |
| 5 | Why do sick children become hyponatraemic on IV fluids? | **Non-osmotic ADH release** (pain, nausea, pneumonia, meningitis, surgery) retains free water; **hypotonic fluid** supplies it. |
| 6 | Which children get restricted (2/3) maintenance? | High-ADH states: **meningitis, pneumonia, bronchiolitis, head injury, post-op, encephalitis** (NICE: 50–80%). AGN/AKI: insensible losses + urine output. |
| 7 | Definition of shock (WHO ETAT 2016) | **All three**: cold extremities, **capillary refill > 3 s**, and a **weak, fast pulse**. |
| 8 | Shock bolus volumes: UCG vs WHO ETAT vs NICE | UCG 2023: **20 mL/kg RL/NS as rapidly as possible**. WHO ETAT 2016: **10–20 mL/kg over 30–60 min**. NICE NG29: **10 mL/kg over < 10 min**. Reassess after each. |
| 9 | What did the FEAST trial show? | In African children (largely Uganda) with severe febrile illness and **impaired perfusion but not hypotension**, fluid **boluses increased mortality**. No boluses unless full shock. |
| 10 | Shock in SAM: fluid and rate | **15 mL/kg over 1 hour** of RL with 5% glucose (or half-strength Darrow's with 5% glucose or 0.45% saline with 5% glucose); repeat once if improving; if not, blood 10 mL/kg over 3 h. |
| 11 | Deficit formula | **Deficit (mL) = % dehydration × weight (kg) × 10.** E.g. 7% of 12 kg = 840 mL. |
| 12 | Neonatal fluid volumes (UCG 2023) | Day 1 **60**, day 2 **90**, day 3 **120**, day 4+ **150 mL/kg/day** (up to 180 once fully fed); 10% glucose first 2 days. |
| 13 | Composition of Ringer's lactate (mmol/L) | Na **130–131**, K **4–5**, Cl **109–111**, lactate **28–29**, Ca **2**; no glucose. |
| 14 | Strength of 15% KCl ampoule and how to make 20 mmol/L | **2 mmol/mL.** Add **10 mL to 1 L** (or 5 mL to 500 mL) = 20 mmol/L; mix well. |
| 15 | Potassium safety limits on a general ward | Max **40 mmol/L** peripherally; acute IV replacement **0.2 mmol/kg/h (max 10 mmol/h)**; only if **passing urine**; **never IV push or IM**. |
| 16 | Oral potassium replacement dose (RCH) | **KCl 1–2 mmol/kg/dose (max 20 mmol/dose)**; max **5 mmol/kg/day** (max 50 mmol/day). Oral route preferred. |
| 17 | Emergency treatment of hyponatraemic seizures (NICE NG29) | **2 mL/kg of 2.7% (or 3%) saline over 10–15 min (max 100 mL)**; repeat once; consider a third dose after checking Na. |
| 18 | Maximum rise of sodium in 24 h when correcting hyponatraemia | **≤ 8 mmol/L/24 h** (Medscape, RCH 6–8); **NICE: never > 12 mmol/L/24 h**. Risk: osmotic demyelination. |
| 19 | Maximum rate of fall of sodium in hypernatraemia | **≤ 0.5 mmol/L per hour** (≈ 10–12 mmol/L per 24 h); replace over **48 h or more**. Risk: cerebral oedema and seizures. |

| # | Question | Answer |
|---|---|---|
| 20 | First drug in hyperkalaemia with ECG changes | **Calcium gluconate 10% 0.5 mL/kg IV (max 20 mL) over 2–5 min**: stabilises the myocardium (does not lower K). |
| 21 | Drugs that shift potassium into cells | **Salbutamol neb** 2.5 mg (≤ 25 kg) / 5 mg; **insulin 0.1 unit/kg + 10% glucose 5 mL/kg**; **sodium bicarbonate 1 mmol/kg** if acidotic. |
| 22 | ECG of hyperkalaemia vs hypokalaemia | **High K**: peaked T, prolonged PR, flat/absent P, wide QRS, sine wave, VF/asystole. **Low K**: flat/inverted T, **U waves**, ST depression, long QT, ectopics. |
| 23 | Causes of hyperkalaemia in Ugandan children | **AKI** (AGN, HUS, severe malaria), **tumour lysis (Burkitt lymphoma)**, **haemolysed heel-prick sample**, haemolysis, salt-wasting **CAH**, transfusion, ACE inhibitors. |
| 24 | Hyponatraemia + hyperkalaemia + hypoglycaemia in a vomiting infant | **Salt-wasting congenital adrenal hyperplasia / adrenal crisis**: saline, **hydrocortisone**, glucose; look for ambiguous genitalia in girls. |
| 25 | Making 0.9% saline with 5% glucose on the ward | Remove **50 mL** from a **500 mL** bag of 0.9% saline and add **50 mL of 50% dextrose**; label the bag. |
| 26 | Drops per minute formula | **Drops/min = volume (mL) × drop factor ÷ time (min).** 20-drop set: mL/h ÷ 3. 60-drop micro-drip: drops/min = mL/h. |
| 27 | Signs of fluid overload in an infant | **Enlarging tender liver**, puffy eyelids, oedema, gallop rhythm, basal crackles, rising RR/falling $SpO_2$, weight gain > 5%. |
| 28 | Monitoring for every child on IV fluids | **Prescription in mL/h**, hourly **fluid chart**, **daily weight**, **Na/K/glucose before and at least every 24 h** (4–6 h if sick), clinical review 4-hourly, switch to oral early. |

SELF-TEST

# Single best answer questions

15 questions. The answers and explanations follow at the end.

**1.** A 3-year-old (14 kg) admitted with bacterial meningitis cannot feed. She is alert, CRT 2 s, normal pulses. Which IV fluid prescription is MOST appropriate for the first 24 hours?

- **A** 5% dextrose at 50 mL/h
- **B** 0.9% saline with 5% glucose at about 33 mL/h
- **C** 0.9% saline at 20 mL/kg over 30 minutes, then maintenance
- **D** Half-strength Darrow's at 1200 mL/day
- **E** 0.9% saline with 5% glucose plus KCl 40 mmol/L at 100 mL/h

**2.** A 4-year-old (16 kg) with severe malaria has fever 39.5 °C, HR 165, cool hands, CRT 2 s and a strong radial pulse. Hb 9 g/dL, BCS 4/5. The ward nurse asks whether to give a 20 mL/kg saline bolus. What is the BEST response?

- **A** Yes, give 320 mL as fast as possible because he is tachycardic
- **B** Give 10 mL/kg albumin instead of saline
- **C** No bolus; start maintenance fluid and reassess within 1 hour
- **D** Give whole blood 20 mL/kg
- **E** Restrict all fluids because of cerebral malaria

**3.** A 2-year-old (11 kg) with pneumonia, on 5% dextrose 1 L every 12 hours for 2 days, has a generalised seizure. RBS 6.8 mmol/L. Na 119 mmol/L. After securing the airway and giving diazepam, what is the MOST appropriate next step?

- **A** Fluid restriction to half maintenance only
- **B** Furosemide 1 mg/kg IV
- **C** Sodium bicarbonate 1 mmol/kg IV
- **D** Correct the sodium to 135 mmol/L over the next 6 hours with 0.9% saline
- **E** 3% saline 22 mL (2 mL/kg) IV over 10–15 minutes, then recheck sodium

**4.** A 9-month-old (7.5 kg) with diarrhoea is extremely irritable with doughy skin. Na 164 mmol/L, K 3.6. He is not shocked but cannot drink well. Which plan is MOST appropriate? Advanced

- **A** NG low-osmolarity ORS or IV 0.9% saline with glucose, replacing the deficit over at least 48 h with Na checks every 2–4 h, aiming to lower Na by no more than 0.5 mmol/L/h
- **B** 5% dextrose at 150 mL/kg/day to bring sodium down quickly
- **C** Water by bottle as much as he wants
- **D** 0.45% saline 20 mL/kg bolus
- **E** Restrict fluids to 2/3 maintenance

**5.** A 6-year-old (20 kg) with Burkitt lymphoma started chemotherapy yesterday. He is oliguric. K 7.4 mmol/L; ECG shows peaked T waves and a widened QRS. What is the FIRST drug to give?

- **A** Calcium polystyrene sulfonate 20 g rectally
- **B** Salbutamol 2.5 mg nebulised
- **C** Calcium gluconate 10% 10 mL IV over 2–5 minutes
- **D** Furosemide 20 mg IV
- **E** Sodium bicarbonate 8.4% 20 mL IV

**6.** A 5-year-old (18 kg) with diarrhoea has K 2.6 mmol/L. He is alert, drinking, passing urine, with no ECG changes. What is the MOST appropriate potassium treatment?

A KCl 15% 5 mL IV push
B IV KCl 60 mmol in 500 mL through a peripheral cannula over 2 hours
C No treatment; it will correct itself
D KCl 1–2 mmol/kg/dose orally (max 20 mmol per dose), recheck K, and check magnesium
E Insulin and glucose

**7.** A 12 kg child is 5% dehydrated (not shocked), cannot take orally after abdominal surgery, and you prescribe deficit plus maintenance over 24 hours. Using a 20 drops/mL giving set, what is the approximate drip rate?

A About 14 drops/min
B About 47 drops/min
C About 70 drops/min
D About 25 drops/min
E About 100 drops/min

**8.** A 3-week-old girl presents with vomiting and poor feeding. She is lethargic. Na 124, K 7.1, glucose 2.4 mmol/L. On examination there is clitoromegaly with fused labia. What is the MOST likely diagnosis? Advanced

A Salt-wasting congenital adrenal hyperplasia
B Pyloric stenosis
C SIADH from meningitis
D Acute gastroenteritis with dehydration
E Nephrotic syndrome

**9.** A 2-year-old (10 kg) with Hb 3.8 g/dL from severe malaria has cold hands, CRT 4 s and a weak, fast pulse. What fluid is MOST appropriate?

A 0.9% saline 20 mL/kg rapidly, then transfuse later
B 5% dextrose 10 mL/kg
C Packed red cells or whole blood as soon as possible, with other IV fluid only to maintain hydration
D Ringer's lactate 40 mL/kg over 1 hour
E No fluids until the Hb is rechecked

**10.** Which of the following is the LEAST appropriate statement about potassium chloride (KCl) for IV use in children?

A 15% KCl contains 2 mmol/mL
B Add potassium only once the child is passing urine
C Mix the bag thoroughly after adding KCl
D On a general ward the maximum acute IV rate is about 0.2 mmol/kg/h
E KCl can be given IM if IV access fails

**11.** A 7-year-old (22 kg) with acute post-streptococcal glomerulonephritis has periorbital oedema, BP 140/95 and passed 200 mL of urine in the last 24 hours. Creatinine 180 µmol/L, K 5.6. Which fluid plan is MOST appropriate? Advanced

A Full Holliday–Segar maintenance IV
B Insensible losses (about 400 mL/m²/day) plus the previous day's urine output, with salt restriction and no added potassium
C 20 mL/kg 0.9% saline bolus to improve urine output
D Ringer's lactate at 2/3 maintenance
E Unrestricted oral fluids because he is thirsty

**12.** You need 10% dextrose for a hypoglycaemic child but the ward only has 50% dextrose and 5% dextrose. How do you prepare about 100 mL of approximately 10% dextrose?

- A 50 mL of 50% dextrose + 50 mL of 5% dextrose
- B 20 mL of 50% dextrose + 80 mL of 5% dextrose
- C Give 50% dextrose undiluted at 1 mL/kg
- D 10 mL of 50% dextrose + 90 mL of 5% dextrose
- E 5 mL of 50% dextrose + 95 mL of sterile water

**13.** A 4-year-old (16 kg) with severe acute malnutrition (MUAC 10.9 cm, oedema +) is lethargic with cold hands, CRT 4 s and a weak, fast pulse. Which fluid order is MOST appropriate? Advanced

- A Ringer's lactate 320 mL as fast as possible, repeat if needed
- B Plan C: 100 mL/kg over 3 hours
- C Ringer's lactate with 5% glucose 240 mL over 1 hour, then reassess
- D ReSoMal 80 mL every 30 minutes orally
- E 0.9% saline 160 mL over 10 minutes

**14.** A well 5-day-old baby has a routine heel-prick sample showing K 6.9 mmol/L. She is feeding well, passing urine, and her ECG is normal. What is the MOST appropriate next step?

- A Give calcium gluconate 10% immediately
- B Start insulin and glucose
- C Give Resonium rectally
- D Repeat potassium on a free-flowing venous sample
- E Arrange peritoneal dialysis

**15.** A 30 kg child is nil by mouth after appendicectomy. Using Holliday–Segar, what is the full daily maintenance volume?

- A 1500 mL
- B 1700 mL
- C 2000 mL
- D 3000 mL
- E 1200 mL

## Answers & explanations

> **1. B** · 0.9% saline with 5% glucose at about 33 mL/h
>
> B is correct: maintenance = 1000 + 200 = 1200 mL/day (50 mL/h); meningitis raises ADH, so give about 2/3 (≈ 800 mL/day ≈ 33 mL/h) of an isotonic fluid with glucose (NICE NG29). A is wrong: plain 5% dextrose is hypotonic and, at full maintenance, risks hyponatraemic seizures. C is wrong: she is not shocked, so no bolus. D is wrong: half-strength Darrow's is hypotonic and the volume is unrestricted. E is wrong: 100 mL/h is twice maintenance, and potassium should not start until urine output is confirmed.

> **2. C** · No bolus; start maintenance fluid and reassess within 1 hour
>
> C is correct: he has only one sign of impaired circulation (cool hands) and a fast pulse from fever, not shock (which needs cold extremities plus CRT > 3 s plus a weak fast pulse). WHO ETAT 2016, after FEAST, advises maintenance fluids and reassessment within an hour. A is wrong: boluses increased mortality in febrile African children without shock (FEAST). B is wrong: albumin boluses also increased mortality in FEAST. D is wrong: Hb 9 g/dL does not need transfusion. E is wrong: he still needs maintenance (glucose-containing) fluid; complete restriction risks hypoglycaemia and dehydration.

**3. E** · 3% saline 22 mL (2 mL/kg) IV over 10–15 minutes, then recheck sodium

E is correct: acute symptomatic hyponatraemia (seizure, Na 119) needs hypertonic saline 2 mL/kg (max 100 mL) over 10–15 min, repeated if symptoms persist (NICE NG29); stop the hypotonic fluid. A is wrong: NICE states hyponatraemic encephalopathy must not be managed by fluid restriction alone. B is wrong: furosemide alone does not correct encephalopathy and may worsen hypovolaemia. C is wrong: bicarbonate has no role. D is wrong: raising Na by 16 mmol/L in 6 h risks osmotic demyelination; after symptoms stop, aim for ≤ 8 mmol/L (NICE max 12) in 24 h.

**4. A** · NG low-osmolarity ORS or IV 0.9% saline with glucose, replacing the deficit over at least 48 h with Na checks every 2–4 h, aiming to lower Na by no more than 0.5 mmol/L/h

A is correct: hypernatraemic dehydration is corrected slowly (≤ 0.5 mmol/L/h, about 10–12 mmol/L/24 h) over ≥ 48 h with isotonic fluid or ORS (NICE allows ORS in hypernatraemia) and frequent sodium checks. B is wrong: a large hypotonic load drops sodium fast and causes cerebral oedema and seizures. C is wrong: free water alone also corrects too fast. D is wrong: boluses are for shock and should be isotonic. E is wrong: he is dehydrated and needs deficit replacement, not restriction.

**5. C** · Calcium gluconate 10% 10 mL IV over 2–5 minutes

C is correct: with ECG changes, the first priority is to protect the myocardium with IV calcium gluconate 10% 0.5 mL/kg (10 mL) over 2–5 minutes; it acts within 3 minutes. Then shift K (salbutamol, insulin + glucose) and remove it. A is wrong: resins act over hours. B is wrong: salbutamol is the right second step but takes 30 minutes and does not protect the heart. D is wrong: he is oliguric (tumour lysis AKI), so furosemide will do little, and it is not first-line. E is wrong: bicarbonate is only for acidosis and should not be first; never give it with calcium in the same line.

**6. D** · KCl 1–2 mmol/kg/dose orally (max 20 mmol per dose), recheck K, and check magnesium

D is correct: the oral route is preferred when the child can drink and there are no ECG changes (RCH): 1–2 mmol/kg/dose (max 20 mmol), repeat after checking K, max 5 mmol/kg/day; check Mg. A is wrong: KCl IV push can cause cardiac arrest; it must always be diluted. B is wrong: 120 mmol/L exceeds the 40 mmol/L peripheral limit and the rate (30 mmol/h) exceeds 0.2 mmol/kg/h (3.6 mmol/h; max 10 mmol/h). C is wrong: K 2.6 is moderate-severe and needs treatment. E is wrong: insulin lowers potassium further.

**7. D** · About 25 drops/min

D is correct: deficit = 5 × 12 × 10 = 600 mL; maintenance = 1000 + 100 = 1100 mL; total 1700 mL/24 h ≈ 71 mL/h; with a 20-drop set, drops/min = 71 × 20 ÷ 60 ≈ 24. (In practice, as a post-op child with high ADH, you would restrict maintenance and check electrolytes.) A is wrong: about 14–15 drops/min (≈ 45 mL/h) replaces maintenance only and ignores the 600 mL deficit. B is wrong: 47 drops/min would be about 140 mL/h, double the requirement. C is wrong: 70 drops/min is the mL/h figure, which equals drops/min only with a 60-drop micro-drip set. E is wrong: 100 drops/min would give 300 mL/h, a dangerous overload.

**8. A** · Salt-wasting congenital adrenal hyperplasia

A is correct: hyponatraemia with hyperkalaemia and hypoglycaemia in a 2–3-week-old with virilised genitalia is classic salt-wasting CAH (21-hydroxylase deficiency): give 0.9% saline, hydrocortisone and glucose, and refer. B is wrong: pyloric stenosis gives hypochloraemic hypokalaemic alkalosis, not hyperkalaemia. C is wrong: SIADH causes euvolaemic hyponatraemia with normal potassium. D is wrong: diarrhoea usually causes low potassium, and it does not explain the genitalia. E is wrong: nephrotic syndrome presents with oedema and proteinuria, rarely in neonates, without hyperkalaemia.

**9. C** · Packed red cells or whole blood as soon as possible, with other IV fluid only to maintain hydration

C is correct: WHO ETAT 2016 recommends that children in shock with severe anaemia (Hb < 5 g/dL) receive blood as early as possible (whole blood 20 mL/kg or packed cells 10 mL/kg) and other fluids only for maintenance. A is wrong: crystalloid boluses dilute an already critical Hb and can precipitate heart failure. B is wrong: dextrose is never a resuscitation fluid. D is wrong: a 40 mL/kg crystalloid load is dangerous in severe anaemia. E is wrong: delaying treatment in shock is fatal; the Hb is already known.

**10. E** · KCl can be given IM if IV access fails

E is the answer (the least appropriate, and dangerous statement): KCl must never be given IM or as an IV push; it causes tissue necrosis and cardiac arrest. Statement A is true: 15% KCl = 150 mg/mL = 2 mmol/mL. Statement B is true: 'no pee, no K', because potassium accumulates in oliguria. Statement C is true: KCl layers at the bottom of an unmixed bag, giving a concentrated bolus. Statement D is true: RCH limits acute replacement on a general ward to 0.2 mmol/kg/h (max 10 mmol/h).

**11. B** · Insensible losses (about 400 mL/m²/day) plus the previous day's urine output, with salt restriction and no added potassium

B is correct: in AGN with oliguria and fluid overload, give insensible losses plus urine output, restrict salt, avoid potassium, and use furosemide and antihypertensives as needed. A is wrong: full maintenance (1540 mL/day) would worsen overload and hypertension. C is wrong: a bolus in an overloaded child risks pulmonary oedema and hypertensive encephalopathy. D is wrong: RL contains potassium and 2/3 maintenance is still too much. E is wrong: unrestricted fluids worsen overload.

**12. D** · 10 mL of 50% dextrose + 90 mL of 5% dextrose

D is correct: 10 mL of 50% (5 g) + 90 mL of 5% (4.5 g) = 9.5 g in 100 mL ≈ 10% (UCG: 1 part 50% with 9 parts 5%). A is wrong: that gives about 27.5%, hyperosmolar and irritant to small veins. B is wrong: that gives 14%. C is wrong: 50% dextrose is sclerosing and causes rebound; dilute it. E is wrong: that gives only 2.5%.

**13. C** · Ringer's lactate with 5% glucose 240 mL over 1 hour, then reassess

C is correct: shock in SAM = 15 mL/kg (240 mL) over 1 hour of a glucose-containing fluid (RL with 5% glucose, half-strength Darrow's with 5% glucose or 0.45% saline with 5% glucose); repeat once if improving, then oral/NG ReSoMal; if not improving, blood 10 mL/kg over 3 h (UCG 2023; WHO ETAT 2016). A is wrong: 20 mL/kg rapid boluses risk heart failure in SAM. B is wrong: Plan C is for well-nourished children. D is wrong: oral ReSoMal is for SAM without shock. E is wrong: a 10 mL/kg rapid bolus (NICE practice) is not the SAM protocol and lacks glucose.

**14. D** · Repeat potassium on a free-flowing venous sample

D is correct: squeezed heel-prick samples are often haemolysed, falsely raising potassium; in a well baby with a normal ECG, repeat on a free-flowing venous sample first (neonatal K also runs slightly higher). A is wrong: calcium is for ECG changes or severe confirmed hyperkalaemia. B is wrong: insulin risks hypoglycaemia in a newborn for a probably spurious result. C is wrong: resins in neonates can cause bowel injury and are not needed here. E is wrong: dialysis is for refractory, confirmed hyperkalaemia with AKI.

**15. B** · 1700 mL

B is correct: 1000 (first 10 kg) + 500 (next 10 kg) + 200 (10 kg × 20) = 1700 mL/day (≈ 70 mL/h); after surgery many children need less (NICE 50–80%). A is wrong: 1500 mL is the value for 20 kg. C is wrong: 2000 mL forgets that above 20 kg the rate falls to 20 mL/kg (UCG's 25 mL/kg variant gives 1750). D is wrong: 3000 mL uses 100 mL/kg for every kilogram. E is wrong: 1200 mL is roughly 2/3 of maintenance, not the full volume.

SELF-TEST

# Clinical cases

Work through each step before reading the model answer.

## Case 1: Seizure on IV fluids: iatrogenic hyponatraemia

You are the intern on the paediatric ward at a regional referral hospital. Musa, a 2-year-old boy (12 kg), was admitted yesterday with severe pneumonia. He has been on '5% dextrose 1 L 12-hourly' since admission and is not feeding. At 06:00 the nurse calls: he has had a generalised seizure lasting 5 minutes.

**1. What are your immediate actions?**

> **ABC**: position, suction, **oxygen** ($SpO_2$ 86% → nasal prongs). If still seizing: **diazepam 0.5 mg/kg rectally** (or 0.3 mg/kg IV slowly). **Check glucose** at the bedside (6.5 mmol/L). **Stop the 5% dextrose.** Send urgent **electrolytes** and creatinine. Call the senior.

**2. How much fluid has he received compared with what he needed?**

> He received **2 L/day of plain 5% dextrose**. Maintenance for 12 kg = 1000 + 100 = **1100 mL/day**, and with pneumonia (SIADH risk) about **2/3 = 730 mL/day**. He got nearly **3 times** the appropriate volume of a **hypotonic** fluid.

**3. Na comes back at 118 mmol/L. He is drowsy and twitching. What do you give?**

> **Hypertonic saline 2 mL/kg = 24 mL of 3% saline (or 2.7%) IV over 10–15 min** (NICE NG29; max 100 mL). Repeat once if symptoms persist; recheck Na and consider a third dose. Expect a rise of about 4–6 mmol/L to stop the seizures.

**4. The twitching stops. Na at 1 hour is 123. Write the ongoing fluid prescription and monitoring.**

> - **0.9% saline with 5% glucose** at **2/3 maintenance ≈ 30 mL/h**; KCl 20 mmol/L once urine output is confirmed
> - **Na every 2–4 h**: total rise **≤ 8 mmol/L in 24 h** after symptoms stop (NICE max 12): target ≤ 126 at 24 h
> - Hourly **fluid chart**, **weigh** now and in 12 h, glucose 4-hourly
> - Start **NG feeds** (expressed breast milk/F-75 not needed; normal milk feeds) as soon as safe and reduce IV volume accordingly

**5. Six hours later he is passing large volumes of dilute urine and Na is 128 (up 10 in 7 h). What is happening and what do you do?**

> As ADH switches off he has a **water diuresis**, and sodium is rising **too fast** (risk of osmotic demyelination). Inform the senior; **slow or stop the saline**, consider **5% glucose** or oral water to slow the rise, and recheck Na in 1–2 h.

**6. What system changes would you propose at the morning meeting?**

> Remove plain 5% dextrose as a default maintenance fluid; ward chart for **mL/h by weight** (4-2-1) and **2/3 rule** for meningitis and pneumonia; ready-made **0.9% saline + 5% glucose** recipe on the wall; **electrolytes at 24 h** for every child on IV fluids; burettes or micro-drip sets for children under 15 kg; report the incident without blame.

**Take-away:** Plain 5% dextrose plus high ADH equals hyponatraemic seizures: use isotonic fluid with glucose at 2/3 maintenance, treat seizures with 2 mL/kg hypertonic saline, and then correct slowly.

## Case 2: Hyperkalaemia in a child with acute kidney injury

Grace, a 6-year-old girl (18 kg), is admitted to Mengo Hospital with 3 days of fever, vomiting and dark ('Coca-Cola') urine after treatment for malaria. She has passed very little urine today. She is drowsy but rousable.

**1. What is your initial assessment and what do you send?**

> **ABC, ETAT**: RR 36, $SpO_2$ 95%, HR 120, BP 118/76, CRT 2 s, not shocked; **pallor**, mild jaundice, no oedema; BCS 4/5. Bladder not palpable. Bedside **RBS** (5.2), **HemoCue** (Hb 6.4), malaria RDT/smear, **urine dipstick** (blood +++). Send **electrolytes, urea, creatinine**, FBC, group and cross-match. Insert a urinary catheter to measure output. Likely **blackwater fever / severe malaria with AKI**.

**2. Results: K 7.0 mmol/L, Na 131, urea 28, creatinine 340 µmol/L, bicarbonate 14. ECG shows peaked T waves. Write your immediate orders.**

> 1. **Stop all potassium** (no RL); cardiac monitor
> 2. **Calcium gluconate 10% 9 mL (0.5 mL/kg) IV over 2–5 min**; repeat if ECG still abnormal
> 3. **Salbutamol 2.5 mg nebulised** (≤ 25 kg), repeat 1–2-hourly
> 4. **Soluble insulin 1.8 units (0.1 unit/kg) IV + 10% glucose 90 mL (5 mL/kg)**; glucose every 30–60 min
> 5. **Sodium bicarbonate 8.4% 18 mL (1 mmol/kg) over 30 min** (she is acidotic), not in the calcium line
> 6. Repeat **K and ECG** in 1–2 h
> 7. Resonium 0.5–1 g/kg 6-hourly orally if K remains high

**3. How will you manage her fluids?**

> She is not shocked and is oliguric: give **insensible losses (about 400 mL/m²/day; BSA ≈ 0.75 m² → 300 mL/day) plus the previous hour's/day's urine output**, as **potassium-free** fluid with glucose (0.9% saline with 5% glucose); weigh 12-hourly; strict fluid chart. Consider **furosemide 1–2 mg/kg IV** once to test for a response. Transfuse cautiously for Hb 6.4 if symptomatic (10 mL/kg packed cells over 3–4 h with monitoring).

**4. Which features would make you call for dialysis?**

> **Refractory hyperkalaemia**, **severe acidosis**, **fluid overload with pulmonary oedema** or uncontrolled hypertension, **uraemic encephalopathy**, rising creatinine with **anuria**. Peritoneal dialysis is available at **Mulago** (and some RRHs): arrange early transfer with a referral letter.

**5. Continue malaria treatment: anything to adjust?**

> **IV artesunate** continues (no renal dose change). Avoid nephrotoxins (NSAIDs, gentamicin unless essential with level/dose adjustment). Review drug doses for AKI. Monitor glucose (malaria and insulin both cause hypoglycaemia).

**6. Present her on the ward round.**

> 'Grace, 6 years, 18 kg, fully immunised, HIV-negative, with severe malaria complicated by **haemoglobinuria and oliguric AKI** (creatinine 340), **hyperkalaemia 7.0 with ECG changes** treated with calcium, salbutamol, insulin–glucose and bicarbonate, **metabolic acidosis** and **anaemia** (Hb 6.4). On restricted K-free fluids (insensible + output), IV artesunate. Plan: K and ECG 2-hourly, discuss dialysis with Mulago if K or creatinine keep rising.'

**Take-away:** Hyperkalaemia with ECG changes: protect the heart with calcium, shift potassium with salbutamol and insulin–glucose, remove it, stop all K, restrict fluids to insensible losses plus output, and refer early for dialysis.

### Case 3: Hypernatraemic dehydration in a formula-fed infant

Baby Ivan, 4 months old, is brought to the emergency unit with 3 days of diarrhoea and fever. His mother returned to work and he is formula-fed; she has been adding extra scoops 'to make him strong'. He weighs 5.4 kg (6.0 kg at the 14-week immunisation visit).

**1. What do you find and how do you interpret it?**

> He is **very irritable**, high-pitched cry, **intensely thirsty**, **doughy skin**, eyes slightly sunken, CRT 2 s, warm hands, HR 170, temperature 38.4 °C. Weight loss 0.6 kg = **10% dehydration**, more than the signs suggest. Suspect **hypernatraemic dehydration** (concentrated feeds + fever + diarrhoea).

**2. Na 158 mmol/L, K 3.2, glucose 7.0. Is he shocked, and what is your fluid strategy?**

> Not shocked (warm hands, CRT 2 s), so **no bolus**. Rehydrate **slowly**: ORS by **NG tube** (or IV 0.9% saline with 5% glucose) replacing the deficit (10% × 5.4 × 10 = 540 mL) plus maintenance (540 mL/day) over **48 h** ≈ 1620 mL/48 h ≈ **34 mL/h**. Aim for a fall in Na of **≤ 0.5 mmol/L/h** (≤ 10–12 mmol/L/24 h).

**3. How will you monitor?**

> **Na 2–4-hourly** initially (RCH: 1–2 h after starting, then 4–6-hourly), glucose, **neurological checks** (irritability, fontanelle, seizures), weight 12-hourly, fluid chart. Add **KCl 20 mmol/L** to IV fluid once passing urine (K 3.2).

**4. At 6 hours Na is 150 (fall of 8 in 6 h). What do you do?**

> That is **1.3 mmol/L/h, too fast**. **Slow the infusion** (or change to a more sodium-rich fluid), inform the senior, recheck Na in 1–2 h, and watch for **seizures** (cerebral oedema). If he convulses: stop hypotonic fluid, give **3% saline 2–3 mL/kg**, glucose check, ICU.

**5. What else does he need?**

> **Zinc 10 mg daily for 10 days** (under 6 months); look for the cause of fever (malaria RDT, urine, ears, chest); continue feeding with **correctly prepared** milk once rehydrating; consider whether he can return to **breastfeeding** (relactation support). Check immunisations (14-week vaccines given).

**6. How do you counsel his mother?**

> Show **correct formula preparation**: boiled water cooled, **1 level scoop per 30 mL** (or per tin instructions), never extra scoops or sugar; cup feeding rather than a bottle; hygiene. Explore **expressing breast milk** for when she is at work, or relactation. Teach ORS for future diarrhoea and **danger signs**.

**Take-away:** Hypernatraemia hides dehydration: weigh the baby, rehydrate over 48 h, measure sodium often, slow down if it falls faster than 0.5 mmol/L/h, and fix the feeding error.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.